PINOUT TABLES - one table per connector, bus plug, splice and end of the set; every pin, the wire on it and where its other end lands

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

ECU-A - Link G4X plug A (centre dash)

housing 4-1437290-1 | contacts 3-1447221-4 | pins per Link G4X XtremeX QSG p.5 pin numbering (1-34)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	w111_c	20	White	🟦🟨🟦🟨	Injector 4 drive (-)	DRB102 receptacle (RH firewall) pin 24	630	ST185-AB-cabin	INJ4
2	w110_c	20	White	🟦🟨🟦🟦	Injector 3 drive (-)	DRB102 receptacle (RH firewall) pin 23	630	ST185-AB-cabin	INJ3
3	w109_c	20	White	🟦🟦⬜⬜	Injector 2 drive (-)	DRB102 receptacle (RH firewall) pin 22	630	ST185-AB-cabin	INJ2
4	w108_c	20	White	🟦🟦⬜🟨	Injector 1 drive (-)	DRB102 receptacle (RH firewall) pin 21	630	ST185-AB-cabin	INJ1
5	w_a5_pin	18	Red	🟦	EFI relay k_efi -> LOGIC SW12 7.5 A fuse (+ IG2 wak...	ECU splice point - SP-A5: ECU main feed (SW12 + IG2 diode)	310	ST185-AB-cabin	SW12
6	w19	20	Orange	⬜🟦⬜	+8 V sensor supply	ECU splice point - SP-8V: +8 V cam sensor supply (ECU A6)	310	ST185-AB-cabin	8V
7	w_drain_ecu_a	20	Black	🟨⬜🟦	Shield drain	6 mm behind the ECU-A plug - SP-SHIELD-A: drains to A7	200	ST185-AB-cabin	SHLD
8	w200_c	20	White	🟨🟦⬜🟦	Crank sensor signal	DRB102 receptacle (RH firewall) pin 74	630	ST185-AB-cabin	CRANKC
9	w_trig2_pin	20	White	🟦🟦🟨	Cam sensor signal	At the ECU plugs - SP-CAM: cam signal + 1.8k pull-up to +8V	310	ST185-AB-cabin	CAM
10	w123_c	20	White	⬜⬜⬜	Coil 4 trigger (IGT)	DRB102 receptacle (RH firewall) pin 12	630	ST185-AB-cabin	COP4
11	w122_c	20	White	⬜🟨⬜	Coil 3 trigger (IGT)	DRB102 receptacle (RH firewall) pin 11	630	ST185-AB-cabin	COP3
12	w121_c	20	White	⬜🟨🟨	Coil 2 trigger (IGT)	DRB102 receptacle (RH firewall) pin 10	630	ST185-AB-cabin	COP2
13	w120_c	20	White	⬜🟨🟨	Coil 1 trigger (IGT)	DRB102 receptacle (RH firewall) pin 9	630	ST185-AB-cabin	COP1
14	w38	20	Blue	🟨⬜⬜	Pedal 1 signal main	accelerator pedal sensor pin 6	1090	ST185-AB-cabin	APS
15	w201_c	20	White	🟦🟦🟦🟦	Coolant temp signal	DRB102 receptacle (RH firewall) pin 70	630	ST185-AB-cabin	ECT
16	w202_c	20	White	🟦🟨🟨🟨	Intake temp signal	DRB102 receptacle (RH firewall) pin 33	630	ST185-AB-cabin	IATC
17	w26_c	20	White	🟦🟨🟦🟨	MAP sensor signal	DRB102 receptacle (RH firewall) pin 17	630	ST185-AB-cabin	MAPC
18	w_ac_kill	20	White	🟦🟨🟦⬜	A/C cancel (ACT)	OEM A/C amplifier pin 11	820	ST185-AB-cabin	ACKILL
19	w_fp85	18	White	⬜⬜🟦	Fuel pump relay (-)	B3-K1 Fuel Pump Relay, TE HCR150 (seated in the battery relay block K1) - ACTIVE-LOW - ...	3130	ST185-AB-cabin	FP
20	w81	20	White	⬜⬜	ETB relay coil (-)	B2-M4 ETB Power Relay (seated in the LOGIC box M4) - ACTIVE-LOW - ECU sinks coil- (85)...	680	ST185-CabinPower	ETB
21	w103_c	20	White	🟦🟦🟨⬜	Boost solenoid (-)	DRB102 receptacle (RH firewall) pin 58	630	ST185-AB-cabin	BOOST
22	w128_c	20	White	🟦🟦⬜	Throttle pos 1 main	DRB102 receptacle (RH firewall) pin 42	630	ST185-AB-cabin	ETBS
23	cab_fspur_l	20	Blue	⬜🟦🟨	FL conditioned out	to front VR conditioner pin 3	530	ST185-AB-cabin	WSSF
24	w68	18	Blue	🟨🟨	Sensor ground	DT06-12 plug (ECU splice point) pin 1	310	ST185-AB-cabin	SGND
25	w70	18	Black	🟨🟨⬜	ECU power ground	ECU splice point - SP-GND-ECU: ECU power ground (A25 A34 B25 B34) to the NEG bar	310	ST185-AB-cabin	GND
26	w_str85	18	White	⬜🟨	Start inhibit (Aux 8)	OEM clutch start switch C11 pin 2 (pedal, starter interlock) pin 2	1090	ST185-AB-cabin	STR
27	w_mrs_c	20	White	🟨🟨⬜⬜	EHPS pump speed	DRB102 receptacle (RH firewall) pin 84	630	ST185-AB-cabin	MRSSPD
28	w76	18	White	🟦🟨	EFI relay coil (-)	B1-R2 EFI Main Relay (seated in the POWER box R2) - ACTIVE-LOW - ECU sinks coil- (85); ...	680	ST185-CabinPower	EFI
29	w_fan85	18	White	🟨⬜	Rad fan relay (-)	B1-R4 Radiator fan relay (seated in the POWER box R4) pin 85	680	ST185-AB-cabin	FAN
30	w16_c	20	White	🟦🟦⬜	Turbo speed signal	DRB102 receptacle (RH firewall) pin 53	630	ST185-AB-cabin	TURBO
31	w48_c	20	White	🟨🟨🟨	Flex fuel signal	DRB102 receptacle (RH firewall) pin 31	630	ST185-AB-cabin	FLEX
32	w67	18	Orange	🟦🟦	+5 V sensor supply	DT06-12 plug (ECU splice point) pin 1	310	ST185-AB-cabin	5V
33	w129_c	20	White	🟦🟦⬜	Throttle pos 2 sub	DRB102 receptacle (RH firewall) pin 44	630	ST185-AB-cabin	ETBS
34	w_a34	18	Black	🟨🟨⬜	ECU power ground	ECU splice point - SP-GND-ECU: ECU power ground (A25 A34 B25 B34) to the NEG bar	310	ST185-AB-cabin	GND

ECU-B (Link G4X plug B, centre dash)

housing 4-1437290-0 | contacts TBD | pins per Link G4X XtremeX QSG p.5 pin numbering (1-34)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1					spare, not used				
2					spare, not used				
3					spare, not used				
4					spare, not used				
5	w82_ac	18	Red	⬜⬜	ETB motor +12 V	B2-M4 ETB Power Relay (seated in the LOGIC box M4) - ACTIVE-LOW - ECU sinks coil- (85)...	680	ST185-CabinPower	ETB
6	w203_c	20	White	🟦🟦🟨🟦	Oil temp signal	DRB102 receptacle (RH firewall) pin 61	630	ST185-AB-cabin	OILT
7	w204_c	20	White	🟦🟨🟨🟦	Charge temp signal	DRB102 receptacle (RH firewall) pin 59	630	ST185-AB-cabin	CIAT
8					spare, not used				
9	w205_c	20	White	🟦🟦🟦⬜	Knock sensor signal	DRB102 receptacle (RH firewall) pin 71	630	ST185-AB-cabin	KNOCK
10	w_tach	20	White	🟨🟦🟨	Engine speed (tacho)	SP-TACH - Ign 8 tacho to A15 pin 18 and 4.7k lock-sensor tap	820	ST185-AB-cabin	TACH
11					spare, not used				
12	w_eps_trig	18	White	🟦⬜	EPS enable relay (-)	B2-M6 EPS Enable Relay (seated in the LOGIC box M6) - ACTIVE-LOW - ECU sinks coil- (85...	680	ST185-AB-cabin	EPSEN
13	w_fan285	18	White	⬜🟦	Cond fan relay (-)	B1-R5 Condenser fan relay (seated in the POWER box R5) pin 85	680	ST185-AB-cabin	FAN2
14					spare, not used				
15	w29_c	20	White	🟦⬜🟦🟦	Oil pressure signal	DRB102 receptacle (RH firewall) pin 28	630	ST185-AB-cabin	OILPC
16	w32_c	20	White	🟦🟦🟦🟨	Fuel pressure signal	DRB102 receptacle (RH firewall) pin 14	630	ST185-AB-cabin	FUELP
17	w_drain_ecu_b	20	Black	🟨⬜🟦	Shield drain B	6 mm behind the ECU-B plug - SP-SHIELD-B: drains to B17	200	ST185-AB-cabin	SHLD
18	w124_c	18	White	🟦🟨⬜	ETB motor +	DRB102 receptacle (RH firewall) pin 45	630	ST185-AB-cabin	ETBM
19	cab_rspur_r	20	Blue	🟦🟦🟨🟨	Rear right wheel spd	IX-WS-R inline to rear VRC (dash above ECU) pin 4	530	ST185-AB-cabin	WSSR
20	cab_rspur_l	20	Green	🟦🟦🟨🟨	Rear left wheel spd	IX-WS-R inline to rear VRC (dash above ECU) pin 3	530	ST185-AB-cabin	WSSR
21	cab_fspur_r_ac	20	White	⬜🟦🟨	FR conditioned out	to front VR conditioner pin 4	530	ST185-AB-cabin	WSSF
22	w69	18	Blue	🟨🟨	Sensor ground B	DT06-12 plug (ECU splice point) pin 1	310	ST185-AB-cabin	SGND
23	w35_c	20	White	⬜🟨🟦	Coolant press signal	DRB102 receptacle (RH firewall) pin 26	630	ST185-AB-cabin	CLNTP
24	w_fl_sig_pin	20	White	🟦🟨🟦	Fuel level signal	SP-FUEL-LVL - fuel sender signal + 470R pull-up to ECU-B B24	310	ST185-AB-cabin	FLVL
25	w_b25_ac	18	Black	🟨🟨⬜	ECU power ground	ECU splice point - SP-GND-ECU: ECU power ground (A25 A34 B25 B34) to the NEG bar	310	ST185-AB-cabin	GND
26	w215_c	18	White	🟦🟨⬜	ETB motor -	DRB102 receptacle (RH firewall) pin 46	630	ST185-AB-cabin	ETBM
27	w_at_immob2	20	White	🟨🟦🟨	Anti-theft input	Alarm module - main harness (DT06-12SA) pin 9	380	ST185-AntiTheft	IMMOB
28	w_ign	20	Yellow	🟦🟦⬜🟦	Ignition sw sense	OEM J/B No.1 (LH kick panel) 1H pin 7, B-O pin 7	1100	ST185-AB-cabin	IGNSW
29	w_vss_sig_c	20	White	🟨🟦🟨	Gearbox speed signal	DRB102 receptacle (RH firewall) pin 68	630	ST185-AB-cabin	VSS
30	w_start	20	Yellow	🟨🟨🟨	Start request	B2 LOGIC box - custom fuse + relay box, glove box inboard (cad/fusebox rev 5, DECISIONS ...	680	ST185-AB-cabin	B30
33	w_aps_vpa2	20	Blue	🟨⬜⬜	Pedal 2 signal sub	accelerator pedal sensor pin 3	1090	ST185-AB-cabin	APS
34	w_b34_ac	18	Black	🟨🟨⬜	ECU power ground	ECU splice point - SP-GND-ECU: ECU power ground (A25 A34 B25 B34) to the NEG bar	310	ST185-AB-cabin	GND

ECU-C - ECU Comms/Tuning Port

housing DTM06-6S + WM-6S | contacts 0462-201-2031 | pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	w_shd_can_trunk	20	Black	🟨	CAN screen trunk	CAN trunk, cluster branch - SP-CAN-SH-CLU: CAN screen	650	ST185-CAN	CAN
2					spare, sealing plug				
3	wc_h_ecu	20	White	🟨	CAN 1 high	CAN H join - ECU, CSB3, cluster, Pi, bulkhead	250	ST185-CAN	CAN
4	wc_l_ecu	20	Green	🟨	CAN 1 low	CAN L join - ECU, CSB3, cluster, Pi, bulkhead	250	ST185-CAN	CAN
5					spare, sealing plug				
6					spare, sealing plug				

BUS-5V - DT06-12 plug (ECU splice point)

housing DT06-12SA + W12S; DT04-12PA-P021 (bussed cap) | contacts 0462-201-16141 | pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	w67	18	Orange	🟦🟦	+5 V from ECU A32	Link G4X plug A (centre dash) pin 32	310	ST185-AB-cabin	5V
2	w_rail_sp_5v_c	20	Orange	🟦🟦	+5 V throttle sensor	DRB102 receptacle (RH firewall) pin 41	520	ST185-AB-cabin	5V
3	w_5v_a_c39	20	Orange	🟦🟦	+5 V MAP sensor	DRB102 receptacle (RH firewall) pin 16	520	ST185-AB-cabin	5V
4	w_5v_a_c40	20	Orange	🟦🟦	+5 V turbo speed	DRB102 receptacle (RH firewall) pin 52	520	ST185-AB-cabin	5V
5	w_5v_brail_c45	20	Orange	🟦🟦	+5 V oil pressure	DRB102 receptacle (RH firewall) pin 27	520	ST185-AB-cabin	5V
6	w_5v_brail_c46	20	Orange	🟦🟦	+5 V fuel pressure	DRB102 receptacle (RH firewall) pin 13	520	ST185-AB-cabin	5V
7	w_5v_brail_c47	20	Orange	🟦🟦	+5 V coolant press	DRB102 receptacle (RH firewall) pin 25	520	ST185-AB-cabin	5V
8	w_aps_vc1	20	Red	🟦🟦	+5 V pedal sensor 1	accelerator pedal sensor pin 4	1200	ST185-AB-cabin	5V
9	w210	20	Red	🟦🟦	+5 V pedal sensor 2	accelerator pedal sensor pin 1	1200	ST185-AB-cabin	5V
10	cab_fspur_5v	20	Red	🟦🟦	+5 V front VRC	to front VR conditioner pin 1	640	ST185-AB-cabin	5V
11	cab_rspur_5v	20	Red	🟦🟦	+5 V rear VRC	IX-WS-R inline to rear VRC (dash above ECU) pin 1	640	ST185-AB-cabin	5V
12	w_5v_bus_link	20	Orange	🟦🟦	+5 V to BUS-5V-2	DT06-8 plug (ECU splice point) pin 1	200	ST185-AB-cabin	5V

BUS-5V-2 - DT06-8 plug (ECU splice point)

housing DT06-08SA + W8S; DT04-08PA-P021 (bussed cap) | contacts 0462-201-16141 | pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	w_5v_bus_link	20	Orange	🟦🟦	+5 V from BUS-5V	DT06-12 plug (ECU splice point) pin 12	200	ST185-AB-cabin	5V
2	w_cru_5v_ac	20	Orange	🟦🟦	+5 V cruise ladder	resistor r_cruise on CSB3 pin 1	680	ST185-AB-cabin	5V
4	w44	20	Orange	🟦🟨🟦	+5 V fuel lvl pullup	470R fuel level pull-up to +5V pin 2	200	ST185-AB-cabin	FLVL
5					spare, sealing plug				
6					spare, sealing plug				
7					spare, sealing plug				
8					spare, sealing plug				

BUS-GNDOUT-A - DT06-12 plug (ECU splice point)

housing DT06-12SA + W12S; DT04-12PA-P021 (bussed cap) | contacts 0462-201-16141 | pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	w68	18	Blue	🟨🟨	Sensor gnd from A24	Link G4X plug A (centre dash) pin 24	310	ST185-AB-cabin	SGND
2	w_rail_sp_gndout_c	20	Blue	🟨🟨	Gnd IAT/TPS/coolant	DRB102 receptacle (RH firewall) pin 34	520	ST185-AB-cabin	SGND
3	w_gnd_a_c41	20	Blue	🟨🟨	Gnd crank sensor	DRB102 receptacle (RH firewall) pin 75	520	ST185-AB-cabin	SGND
4	w_gnd_a_c42	20	Blue	🟨🟨	Gnd cam sensor	DRB102 receptacle (RH firewall) pin 65	520	ST185-AB-cabin	SGND
5	w_gnd_a_c43	20	Blue	🟨🟨	Gnd turbo speed	DRB102 receptacle (RH firewall) pin 56	520	ST185-AB-cabin	SGND
6	w_gnd_a_c44	20	Blue	🟨🟨	Gnd flex + MAP	DRB102 receptacle (RH firewall) pin 18	520	ST185-AB-cabin	SGND
7	w40	20	Yellow	🟨🟨	Pedal 1 ground	accelerator pedal sensor pin 5	1200	ST185-AB-cabin	SGND
8	w_aps_gnd2	20	Yellow	🟨🟨	Pedal 2 ground	accelerator pedal sensor pin 2	1200	ST185-AB-cabin	SGND
9	cab_fspur_gnd	20	Yellow	🟨🟨	Front VRC ground	to front VR conditioner pin 2	640	ST185-AB-cabin	SGND
10					spare, sealing plug				
11					spare, sealing plug				
12					spare, sealing plug				

BUS-GNDOUT-B - DT06-12 plug (ECU splice point)

housing DT06-12SA + W12S; DT04-12PA-P021 (bussed cap) | contacts 0462-201-16141 | pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	w69	18	Blue	🟨🟨	Sensor gnd from B22	ECU-B (Link G4X plug B, centre dash) pin 22	310	ST185-AB-cabin	SGND
2	w45	20	Blue	🟨🟨	Fuel level ground	IX-FUEL inline to fuel sender lead (trunk end) pin 2	2820	ST185-AB-cabin	SGND
3	cab_rspur_gnd	20	Yellow	🟨🟨	Rear VRC ground	IX-WS-R inline to rear VRC (dash above ECU) pin 2	640	ST185-AB-cabin	SGND
6	w12_c	20	Blue	🟦🟦🟦⬜	Knock sensor (-)	DRB102 receptacle (RH firewall) pin 72	520	ST185-AB-cabin	KNOCK
7	w_gnd_b_c19	20	Blue	🟨🟨	Gnd oil/clnt/temps	DRB102 receptacle (RH firewall) pin 29	520	ST185-AB-cabin	SGND
8	w_gnd_b_c21	20	Blue	🟨🟨	Gnd fuel press + VSS	DRB102 receptacle (RH firewall) pin 15	520	ST185-AB-cabin	SGND
9					spare, sealing plug				
10					spare, sealing plug				
11					spare, sealing plug				
12					spare, sealing plug				

BUS-SW12 - SW12 bus plug DT06-12 (glove box, by LOGIC box)

housing DT06-12SA + W12S; DT04-12PA-P021 (bussed cap) | contacts 0462-201-16141 | pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	w_cp_sw12_in	16	Red	🟦	EFI relay k_efi -> LOGIC SW12 7.5 A fuse	B2 LOGIC box - custom fuse + relay box, glove box inboard (cad/fusebox rev 5, DECISIONS ...	200	ST185-CabinPower	SW12
2	w_fb_sw12	18	Red	🟦	EFI relay k_efi -> LOGIC SW12 7.5 A fuse	ECU splice point - SP-SW12: switched +12 V (LOGIC SW12)	570	ST185-AB-cabin	SW12
3	w_k_fp_86_cp	18	Red	🟦	EFI relay k_efi -> LOGIC SW12 7.5 A fuse	B3-K1 Fuel Pump Relay, TE HCR150 (seated in the battery relay block K1) - ACTIVE-LOW - ...	2650	ST185-CabinPower	SW12
4	w_k_fan_86_cp	18	Red	🟦	EFI relay k_efi -> LOGIC SW12 7.5 A fuse	B1-R4 Radiator fan relay (seated in the POWER box R4) pin 86	200	ST185-CabinPower	SW12
5	w_k_etb_86_cp	20	Red	🟦	EFI relay k_efi -> LOGIC SW12 7.5 A fuse	B2-M4 ETB Power Relay (seated in the LOGIC box M4) - ACTIVE-LOW - ECU sinks coil- (85)...	200	ST185-CabinPower	SW12
7	w_k_fan2_86_cp	18	Red	🟦	EFI relay k_efi -> LOGIC SW12 7.5 A fuse	B1-R5 Condenser fan relay (seated in the POWER box R5) pin 86	200	ST185-CabinPower	SW12
8	w_k_eps_86_cp	18	Red	🟦	EFI relay k_efi -> LOGIC SW12 7.5 A fuse	B4 EPS pump relay (HCR150, glove box, beside the POWER box) - coil- (85) switched to gro...	200	ST185-CabinPower	SW12
9	w_k_eps_en_86_cp	18	Red	🟦	EFI relay k_efi -> LOGIC SW12 7.5 A fuse	B2-M6 EPS Enable Relay (seated in the LOGIC box M6) - ACTIVE-LOW - ECU sinks coil- (85...	200	ST185-CabinPower	SW12
10					spare, sealing plug				
11					spare, sealing plug				
12					spare, sealing plug				

SP-12V-SHK - Shock sensor +12 V splice - LOGIC ALMS 2A fuse to shock sensor pin A

splice: solder splice

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_at_12v_shk_tie_cp	20	Red	🟨⬜⬜	AMI BUS1 50 A fuse -> LOGIC ALMS 2 A fuse	B2 LOGIC box - custom fuse + relay box, glove box inboard (cad/fusebox rev 5, DECISIONS ...	860	ST185-CabinPower	ALMS

REVISIONS

REV	DATE	BY	DESCRIPTION
A	2026-10-20	DG	First issue: every harness of the master in one drawing set (replaces the ST185-AB-cabin pilot, R...
B	2026-10-25	DG	Surgical rewire written into the master: 21 decided circuits, D16 A/C, OEM starter / reverse / rear l...
C	2026-10-26	DG	A15 lock-sensor input takes a modified tacho: Ign 8 through 4.7 k· to pin 9 (N61 workaround).
D	2026-10-26	DG	Car-check answers: if the ignition-switch black-orange wire is at the engine plug, remove it; A/C s...

ST185-harness - every wiring harness of the car, one drawing set

Pinout tables - every pin of every connector, bus, splice and end

DRAWING NO.	REV	SHEET	SIZE	DRAWN	DATE
ST185-harness-PIN	**D**	**1 OF 1**	**1716 x 1403 mm**	**DG**	**2026-10-26**

ST185 Celica (living-relation) - generated by docs/harness/tools/build_drawings.py from docs/harness/MASTER-wirelist.csv; not to scale